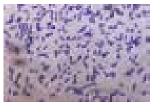
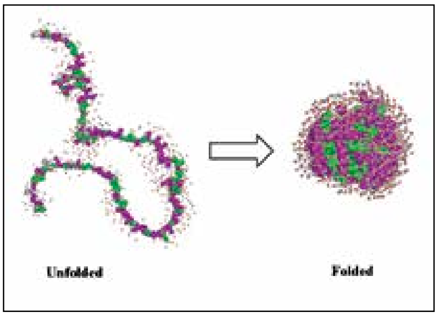
Car turns into motorcycles

The Lane Splitter concept car can be split into a pair of motorcycles and also function together as a car. **http://dgit.in/LaneSplitCar**



Fun Fact

The strongest creatures on Earth are gonorrhea bacteria. They can pull 100,000 times their own body weight.



A simple schematic of a protein in a folded and unfolded state

the specific action of the protein. How a protein folds and assumes a specific functional shape is a long-standing problem in science. It's even been declared by Science magazine as one of the biggest unsolved problems in science. The problem in essence includes three parts: 1. How exactly does a protein evolve into its final native structure? 2. Can we come up with a computational algorithm to predict the structure of a protein given its amino acid sequence? 3. Given the large number of possible conformations how exactly does a protein fold so quickly? Over the past few decades, significant progress has been made on all three fronts however, scientists are still to completely decode the driving mechanisms and underlying principles governing protein folding.

During the process of folding, a large number of forces and interactions are at play, driving a protein to reach a state of lowest free energy possible which lends it stability. Due to the sheer complexity of the structure and the large number of force fields involved, it's difficult to understand the precise physics of a folding process beyond small proteins. The problem of structure prediction has been attacked by a combination of physics and powerful computers. While they've achieved success with small and relatively simple proteins, scientists still struggle with accurately predicting the folded shape of more complex multidomain proteins from its amino acid sequence, known as a 'priori'.

To understand the process, imagine being at a crossroad with a 1,000 different roads going to the same destination and you choosing the path that took the least amount of time to get there. A similar, but larger scale problem is the kinetic mechanism of a protein folding to a specific state amidst many possible structures. It has been figured out that random thermal motions play a major role in the rapid nature of folding and that the protein 'zips' through conformations locally, avoiding unfavourable structures, the physical route is an open question – solving which could also lead to faster algorithms for protein structure prediction.

The protein folding problem is arguably the hottest topic today in biochemical and biophysics research. The physics and computing algorithms developed for protein folding have led to the development of new man-made polymeric materials. Besides contributing to the growth of scientific computing, the problem has resulted in a better understanding of diseases like Type II Diabetes, Alzheimer's, Parkinson's and Huntington's – diseases in which protein misfolding plays a part. A better physical understanding of protein folding could not only lead to several breakthroughs in materials and biological sciences but could also revolutionise medicine.

DID YOU KNOW?

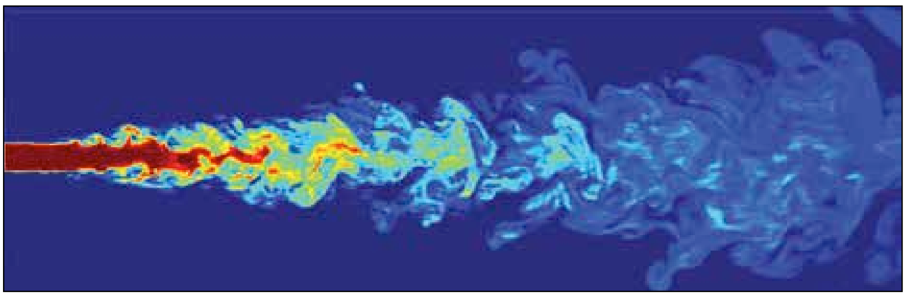
Investment management firm D.E Shaw's founder, David Shaw, also founded D.E Shaw Research, the company that built a computer called 'Anton'. The computer is one of the most advanced resources for running high fidelity simulations for protein physics.

A Quantum Theory of Gravity

We all know that an apple supposedly fell on Newton's head and led to the discovery of gravity. Suffice to say, the world wasn't the same anymore. Then came along Albert Einstein with his celebrated theory of general relativity. He re-imagined gravity as the distortion of space-time, the fabric that makes up the universe. Imagine a heavy ball on a stretched bedsheet and another smaller ball a little further away from it. The big ball would press down on the sheet depressing it, causing the smaller ball to be drawn towards it. Einstein's theory of gravity worked wonders and even explained the bending of light. However, when it came to sub-atomic particles governed by the law of quantum mechanics, the theory threw up all sorts of weird results. Developing a theory of gravity that can unify quantum mechanics and relativity, two of 20th century's most successful theories, has become the biggest research topic ever since.

This problem has birthed some new and exciting areas in physics and mathematics. The area that has gained the most traction is 'String theory'. String theory replaces the notion of particles with tiny vibrating strings that can assume various shapes. Each string can vibrate in a particular mode lending it a specific mass and spin. String theory is incredibly complicated and mathematically deals with ten dimensions in space-time – six more than what humans can physically perceive. The theory has been successful in explaining many of the oddities of gravity's marriage with quantum mechanics and has been hailed as one of the top contenders for a 'theory of everything'.

Another theory for formulating quantum gravity is 'loop quantum gravity' (LQG). LQG is relatively less ambitious, in the sense that it primarily aims to be a consistent theory of gravity without trying to achieve any grand unification of sorts.



SOURCE: TASC.LLNL.GOV

A simulation snapshot showing turbulence in a jet